

Assessing Critical Thinking through a Multi-Agent LLM-Based Debate Chatbot

Bogyeom Park
Department of Applied Artificial Intelligence
Seoul National University of Science and Technology
Seoul, Republic of Korea
bogyeom@seoultech.ac.kr

Kyoungwon Seo
Department of Applied Artificial Intelligence
Seoul National University of Science and Technology
Seoul, Republic of Korea
kwseo@seoultech.ac.kr

Stage 1 Establishing evidence		Stage 2 Adding a warrant	
Dialogue with the argumentation chatbot	Evaluation framework	Dialogue with the argumentation chatbot	Evaluation framework
Great. First, could you provide one supporting reason for our argument? Algorithms can <u>limit users' choices and provide biased information, thereby reducing the diversity of thought.</u> Now, could you present a second reason? Algorithms can often be <u>manipulated for commercial interests, suggesting products that are irrelevant to the user's actual needs or encouraging overconsumption.</u>	Argumentation skills: High (7-9) <u>Two supporting reasons</u> were provided to substantiate the argument. Intellectual standards: "-" (7) The claim that algorithms <u>restrict choices and reduce diversity</u> lacks accurate evidence, leading to lower score. ⇒ Result: High- (7)	Can you provide an example of algorithms reducing diversity through bias? There was a case where <u>YouTube's algorithm repeatedly recommended the same type of content based on viewing history</u>, causing users to miss opportunities to encounter new information. Can you give an example of algorithms driving overconsumption for profit? <u>Some e-commerce platforms have been criticized for prominently displaying certain products at the top</u>, causing consumers to make unnecessary purchases.	Argumentation skills: High (7-9) <u>Two examples supporting</u> the reasons were provided. Intellectual standards: "0" (8) If specific examples like YouTube's suggestions are expanded with broader context, the argument could be even stronger. ⇒ Result: High0 (8)
Stage 3 Presenting a rebuttal		Stage 4 Proposing a counter-rebuttal	
Dialogue with the argumentation chatbot	Evaluation framework	Dialogue with the argumentation chatbot	Evaluation framework
Algorithm suggestions may infringe on privacy and threaten data security by risking the leakage or misuse of personal information during data collection. I think these concerns can be addressed by <u>enforcing strict security measures and implementing laws</u> that require companies to rigorously prevent data leaks. Algorithm suggestions homogenize preferences, reducing diversity and limiting discovery of new perspectives. I believe that algorithm recommendations can <u>offer unique experiences by providing suggestions based on the user's interests.</u>	Argumentation skills: Middle (4-6) <u>Only evaluates the opponent's argument logically once</u> and points out its flaws. Intellectual standards: "0" (5) The rebuttal lacks diverse perspectives, reducing its ability to logically counter the opponent's argument. ⇒ Result: Middle0 (5)	Regarding your reason of "convenience," can we say that the content suggested by algorithms is always optimal? If users regularly input incorrect data or if their preferences change, isn't there a possibility of continuously receiving incorrect recommendations? Of course, <u>algorithms can't be perfect</u>. However, <u>recent technologies continuously adapt to user preferences</u>. Moreover, algorithms don't make all the decisions; choices remain with the users, so I don't think it's a major issue.	Argumentation skills: High (7-9) User countered the opponent's <u>rebuttals twice</u> to strengthen their own logic. Intellectual standards: "-" (7) User refuted the opponent's points specifically, but the argument lacks broader implications of algorithmic design flaws. ⇒ Result: High- (7)

Figure 1: Assessment of students' critical thinking skills. Underlined sections indicate logical elements evaluated for Argumentation Skills, while highlighted sections represent areas contributing to Intellectual Standards assessments.

Abstract

Critical thinking (CT) is a crucial competency in education, requiring structured and reliable assessment methods. This study introduces an LLM-based debate chatbot framework designed to evaluate CT by integrating argument construction and analytical reasoning (i.e., argumentation skills) and qualitative reasoning criteria (i.e., intellectual standards). Two assessment models were developed: the Single-Agent (SA) model and the Multi-Agent (MA) model. The MA model achieved an Intraclass Correlation Coefficient (ICC) of 0.78 and 97.37 % Agreement within ± 1 with human evaluators, indicating strong alignment with human assessment results. By independently assessing argumentation skills and intellectual standards, the MA model more effectively mirrored

the nuanced assessment patterns observed in human assessments. These findings highlight the potential of multi-agent approaches in CT assessment, bridging structured assessment models with expert-like evaluative reasoning. This study contributes to the development of LLM-driven CT assessment frameworks, offering a scalable foundation for integrating diverse evaluative criteria in automated educational assessment.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); Interaction design; • Applied computing → Education;

Keywords

Critical thinking assessment, Debate chatbot, Multi-Agent, Large language model

ACM Reference Format:

Bogyeom Park and Kyoungwon Seo. 2025. Assessing Critical Thinking through a Multi-Agent LLM-Based Debate Chatbot. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA*

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3721207>

'25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 13 pages. <https://doi.org/10.1145/3706599.3721207>

1 Introduction

Critical Thinking (CT) is a core competency encompassing logical analysis and problem-solving abilities, and considerable research has been dedicated to developing effective methods for its assessment [7, 10]. However, conventional assessment tools such as standardized tests and written assignments primarily capture isolated aspects of CT (e.g., logical reasoning, inferential skills), failing to provide a comprehensive assessment of learners' cognitive processes [8]. To overcome these limitations, debate has emerged as a promising alternative for directly assessing CT through structured argumentation and rebuttals [1, 6]. Empirical studies suggest that engaging in debates—where learners construct evidence-based arguments and critically examine opposing perspectives—enhances both logical reasoning and problem-solving skills [15]. Given these advantages, the integration of debate as a CT assessment tool is gaining traction in both educational and research contexts, with ongoing efforts to refine and advance methodologies for more sophisticated CT assessment.

Traditionally, instructors have assessed students' CT in debate settings by considering both argumentation skills and intellectual standards. This involves evaluating the logical structure of students' claims, evidence, and rebuttals to assess the coherence and persuasiveness of their reasoning [2, 19], while also incorporating qualitative dimensions of thinking—such as clarity, fairness, and relevance—to gauge the depth and balance of their cognitive processes [4, 13]. However, this conventional assessment approach presents significant challenges. It requires instructors to individually analyze each student's debate performance, imposing substantial time and labor demands [5], while also introducing a high degree of subjectivity, potentially compromising the reliability of assessments [16]. To address these limitations, recent research has focused on enhancing the objectivity and reliability of CT assessments, with particular attention given to the development of automated assessment techniques.

Building on this line of research, LLM-based CT assessment tools have recently gained significant attention [17]. For instance, DEBO, a Single-Agent (SA) model leveraging ChatGPT, analyzes students' debates by assessing their logical structure (i.e., claims, evidence, and rebuttals) and providing feedback [9]. While SA models effectively assess logical reasoning, they struggle to evaluate qualitative dimensions such as clarity, fairness, and depth.

To enhance the reliability and precision of CT assessment, Multi-Agent (MA) models have emerged as a promising alternative. Recent studies suggest that MA models not only assess the logical structure of texts but also replicate human assessors' collaborative assessment process through agent interactions [3, 11]. This architecture facilitates a more nuanced analysis and marks a significant technological advancement in automating and enhancing CT assessment reliability.

This study aims to develop a multidimensional CT assessment framework by integrating logical and qualitative assessment through an MA model. By implementing this framework in an LLM-based

chatbot, the study seeks to establish a more precise and reliable assessment system. Specifically, we:

- (1) Design a CT assessment framework that balances logical and qualitative dimensions.
- (2) Compare the reliability and validity of SA and MA models using the Intraclass Correlation Coefficient (ICC) and Agreement within ± 1 as assessment metrics.

This research contributes to enhancing the reliability and fairness of automated CT assessment while providing a structured foundation for critical thinking education. By reinforcing objectivity in CT assessment and improving the practicality of LLM-based AI systems in debate education, this study is expected to play a significant role in advancing AI-assisted learning methodologies.

2 MATERIALS AND METHOD

2.1 Participants

A total of 20 university students (10 males, 10 females; mean age = 27.15 ± 1.81 years) were recruited through a university community. Participants were randomly assigned to either the pro ($n = 10$) or con ($n = 10$) position on the debate. Each group consisted of 5 male and 5 female participants. All participants were informed in advance that they could withdraw from the study at any time if they felt uncomfortable during the experimental process; however, all participants completed the study. Written consent was obtained from all participants prior to the study, which was conducted in accordance with ethical guidelines approved by the Institutional Review Board at Seoul National University of Science and Technology (IRB approval number: 2023-0021-01).

2.2 Chatbot design and structure

The debate chatbot was designed to structure the debate process into four distinct stages: (1) Establishing Evidence, (2) Adding a Warrant, (3) Presenting a Rebuttal, and (4) Proposing a Counter-Rebuttal (see Table 1). This four-stage framework is based on Toulmin's model of argumentation, which defines structured reasoning through claims, evidence, warrants, and rebuttals as essential components of argumentation [18].

At each stage, the chatbot presented predefined prompts and follow-up questions, encouraging participants to refine and elaborate on their reasoning. If a student's response contained fewer than 20 characters or was left blank, the chatbot provided predefined feedback prompting them to reconsider and refine their response. This adaptive design enhances student engagement while fostering autonomous reasoning, which is a critical component of CT development [20].

To ensure consistency in data collection and maintain a structured assessment process, the debate chatbot was designed around the topic *"Does algorithmic recommendation make our lives more useful?"* This topic was chosen because it represents a contemporary technological and ethical issue widely debated in modern discourse. Engaging in debates on real-world issues has been demonstrated to enhance students' ability to evaluate evidence, recognize multiple perspectives, and construct well-reasoned arguments [12]. Through this structured debate framework, students not only strengthen

Table 1: Argumentation chatbot with four stages

Stage	Criteria
Establishing Evidence	Assessing the student's ability to present clear and logical evidence to support their claim.
Adding a Warrant	Evaluating the student's ability to analyze their evidence, add specific and reliable data, and strengthen their reasoning.
Presenting a Rebuttal	Measuring the student's ability to critically evaluate the opposing argument, identify flaws, and expand the discussion.
Proposing a Counter-Rebuttal	Assessing the student's ability to respond to rebuttals, reflect on their reasoning, and enhance their argument.

their argumentation skills but also adopt a more systematic approach to analyzing complex socio-technological dilemmas. For a detailed example of the chatbot's interaction with a student, refer to Appendix A.

2.3 Assessment framework

Upon completion of the debate process, students' CT is assessed using the Integrative Argumentation Assessment Framework proposed in this study. This framework integrates argumentation skills and intellectual standards to provide a balanced assessment of students' reasoning. The argumentation skills outlined by Cargas et al. [2] assess how well students construct, justify, and defend their arguments, while the intellectual standards defined by Richard Paul and Linda Elder [13] ensure that their reasoning meets key assessment criteria such as clarity, accuracy, and relevance.

To enhance reliability and fairness, this study employs a dual-layered scoring approach [14], integrating quantitative assessment (i.e., argumentation skills) with qualitative refinement (i.e., intellectual standards). By incorporating a structured scoring mechanism, this approach improves assessment consistency and objectivity. Closely aligned with the chatbot's four-stage structure, the framework systematically maps each stage to corresponding argumentation skills and intellectual standards, ensuring a comprehensive and structured CT assessment process.

2.3.1 Evaluating argumentation skills. Cargas et al. (2017) conceptualized CT as a multidimensional competency comprising various argumentation skills, including hypothesizing, making claims, analyzing, investigating, evaluating, synthesizing, integrating, and reflecting [2]. Among these, four key skills—Make Claims, Analyze, Evaluate, and Reflect—were selected as they directly correspond to the structured stages of debate implemented in this study. These skills align with the core phases of argumentation, progressing from formulating claims to analyzing evidence, evaluating opposing arguments, and refining reasoning.

To ensure a structured assessment aligned with the chatbot's debate process, these four skills were mapped to the corresponding debate stages (see Table 2). As the foundation of the dual-layered scoring approach, argumentation skills provide an initial quantitative assessment of students' reasoning. This mapping clarifies how argumentation skills develop progressively throughout the debate process, reinforcing logical reasoning at each stage. For example, "Make Claims" is critical in the "Establishing Evidence" stage, where students present clear and logical arguments supported by evidence.

Through this mapping, students systematically develop their argumentation skills, progressing from constructing initial claims to critically evaluating and refining their reasoning throughout the debate process.

2.3.2 Evaluating intellectual standards. To assess the qualitative aspects of CT, this framework integrates eight intellectual standards proposed by Paul and Elder (2013) [13]. Among these, clarity is considered a prerequisite for all other standards and is therefore not evaluated independently. Since the chatbot debate process follows a structured four-stage model, four intellectual standards were selected to align with each stage. These standards—Relevance and Accuracy, Depth and Significance, Logic and Breadth, and Fairness and Precision—were chosen for their direct applicability to argumentation quality in a debate setting, ensuring that both factual reliability and logical coherence are maintained throughout the discussion.

To systematically assess both logical reasoning and qualitative aspects of argumentation, each stage of the chatbot interaction is aligned with a corresponding intellectual standard (see Table 3). These intellectual standards constitute the second layer of the dual-layered scoring approach, refining the initial score based on qualitative reasoning depth. By integrating argumentation skills and intellectual standards, this framework provides a structured and comprehensive CT assessment, reinforcing both logical structure and qualitative depth in students' reasoning.

2.3.3 Score augmentation method. To enhance the reliability and fairness of CT assessment, this study adopts the dual-layered scoring approach, integrating argumentation skills with intellectual standards. This approach addresses issues arising from overly complex or ambiguous assessment criteria, which can cause inconsistencies in evaluator interpretation and reduced reliability [14]. By reinforcing consistency in assessment, this method mitigates challenges in CT assessment, such as subjectivity and score variability.

First, each student response is assigned a preliminary score (1 to 9) based on how well it aligns with the argumentation skill relevant to that stage. The scoring is categorized as follows:

- **High (7-9):** Responses fully met the stage requirements, demonstrated well-developed reasoning, and included at least two well-constructed statements or examples.
- **Middle (4-6):** Responses partially fulfilled the requirements, showed some reasoning but lacked depth or completeness, or included only one well-constructed statement.

Table 2: Mapping argumentation skills to the debate chatbot stages

Stage	Argumentation Skills	Definition
Establishing Evidence	Make Claims	Articulating clear and logical reasons supporting the argument.
Adding a Warrant	Analyze	Enhancing reasoning with data and critical analysis.
Presenting a Rebuttal	Evaluate	Assessing and challenging opposing arguments.
Proposing a Counter-Rebuttal	Reflect	Reflecting on feedback to improve argument quality.

Table 3: Mapping intellectual standards to the debate chatbot stages

Stage	Intellectual Standards	Definition
Establishing Evidence	Relevance and Accuracy	Presenting topic-relevant and factually accurate evidence.
Adding a Warrant	Depth and Significance	Highlighting complexity and importance in examples.
Presenting a Rebuttal	Logic and Breadth	Offering logically sound and diverse perspectives.
Proposing a Counter-Rebuttal	Fairness and Precision	Responding impartially and with precise reasoning.

- **Low (1-3):** Responses failed to meet the requirements, demonstrated minimal reasoning, or were absent.

Second, a secondary adjustment is made based on two intellectual standards per stage (see Table 3). This adjustment ensures that the qualitative depth of reasoning is also reflected in the final score:

- **“+” (Upper bound):** Both intellectual standards were fully met, leading to the maximum possible score for that category.
- **“0” (Midpoint):** One or both intellectual standards were partially met, resulting in a moderate adjustment.
- **“-” (Lower bound):** Neither intellectual standard was met, reducing the score to the minimum possible for that category.

To enhance fairness, the final scores are averaged across three independent assessments, reducing evaluator bias and ensuring balanced assessment. Additionally, the framework provides detailed feedback, helping students identify strengths and areas for improvement in critical thinking. A detailed score calculation process is illustrated in Figure 1.

2.3.4 Comparing single-agent and multi-agent models. In this study, we designed and compared a SA model and MA model to evaluate reliability and agreement with human assessments. The SA model employed a simplified structure, where a single agent was responsible for assessing both argumentation skills and intellectual standards. In contrast, the MA model separated these tasks, assigning one agent to assess argumentation skills and another to evaluate intellectual standards. This separation was intended to reduce cognitive overload on a single agent and enhance assessment consistency. The assessment results of both models were systematically compared against human assessments, focusing on how structural differences influenced reliability and agreement. To ensure transparency in scoring criteria and decision-making, both models explicitly defined the dual-layered scoring approach, with the MA model aligning more closely with this structure by assigning distinct agents for argumentation skills and intellectual standards assessment. Detailed descriptions of the prompts, including specific instructions for assessing argumentation skills and intellectual standards, are provided in Appendix A.

2.4 Experimental Procedure

Participants were introduced to the debate topic, “*Does algorithmic recommendation make our lives more useful?*”, through pre-designed reading materials, which they could review before engaging with the chatbot. To maintain controlled conditions, external references such as internet searches were prohibited. Before the debate, participants briefly interacted with the chatbot, exchanging simple greetings. Once ready, they were randomly assigned to either the pro or con position.

The debate followed a structured four-stage process: (1) Establishing Evidence, (2) Adding a Warrant, (3) Presenting a Rebuttal, and (4) Proposing a Counter-Rebuttal. At each stage, the chatbot presented predefined prompts and follow-up questions to guide participants in constructing their arguments. For example, if a participant was assigned the con position, the chatbot would prompt them to provide evidence supporting their stance. If a response was under 20 characters or lacked substantive content (e.g., “*I don’t know*”), the chatbot provided predefined feedback encouraging elaboration (e.g., “*Would you like to reconsider your response?*”).

All participant responses were logged and stored for experiment analysis. By the end of the debate, each participant had generated four distinct responses, corresponding to the four debate stages. These responses were later assessed using the Integrative Argumentation Assessment Framework, which applied the dual-layered scoring approach to evaluate both argumentation skills and intellectual standards. Each debate session lasted approximately 20 minutes per participant, and all interactions were systematically recorded to ensure data consistency and reliability.

2.5 Data analysis

To assess whether the chatbot’s scoring aligns with human evaluators and can be considered reliable, two statistical analyses were conducted: ICC and Agreement within ± 1 .

- **ICC** assessed the *relative reliability*, measuring the consistency of score rankings and variance between the chatbot models and human evaluators.

- **Agreement within ± 1** examined the *absolute agreement*, representing the proportion of scores where differences fell within a one-point range.

For human assessments, three independent researchers evaluated all responses using the dual-layered scoring approach, incorporating both argumentation skills and intellectual standards. Each researcher followed a two-step scoring process:

- (1) Assigned a preliminary score (1–9) based on argumentation skills.
- (2) Adjusted the score according to intellectual standards (i.e., “+”, “0”, or “−”).

To ensure consistency, if at least two researchers assigned the same score, that score was retained. If all three scores differed, researchers conducted a consensus discussion to determine a final agreed-upon score.

Then, the SA and MA models also utilized the dual-layered scoring approach through predefined prompts. Each model independently assessed student responses three times, and the final score was determined by averaging the three assessments. These assessments provided a baseline for comparing the chatbot’s SA and MA models against human scoring standards, ensuring a rigorous assessment of their alignment with human evaluators.

3 Results

3.1 Differences in single-agent and multi-agent in ICC

To compare the relative reliability of the SA and MA models with human assessments, ICC was calculated. The ICC between the SA model and human assessments was 0.09, indicating very low consistency in score variability. This suggests that the SA model failed to reflect the relative patterns observed in human scoring, demonstrating its inability to maintain consistent rankings and variability. In contrast, the ICC between the MA model and human assessments was 0.78, indicating a high level of reliability. The MA model produced score patterns closely aligned with human evaluators, reducing scoring inconsistency and evaluator bias. These results highlight the effectiveness of the MA model in capturing both logical reasoning and qualitative depth, making it a more reliable tool for CT assessment.

3.2 Differences in single-agent and multi-agent in agreement within ± 1

To evaluate the absolute agreement between CT assessment models and human assessments, Agreement within ± 1 was calculated. The SA model achieved an Agreement within ± 1 of only 13.16%, with most scores falling outside the ± 1 range. This indicates that the SA model struggled to produce scores that matched human assessments, highlighting its difficulty in aligning with human scoring patterns in terms of absolute differences. In contrast, the MA model achieved an Agreement within ± 1 of 97.37%, with the majority of scores falling within the ± 1 range. This demonstrates the MA model’s ability to minimize absolute discrepancies, ensuring higher alignment with human evaluators. By independently incorporating

argumentation skills and intellectual standards, the MA model significantly improved absolute agreement, making it a more practical and reliable approach for CT assessment tasks.

4 Discussion

This study examined the effectiveness of a MA model compared to a SA model in assessing CT through a debate chatbot. The findings demonstrate that the MA model significantly outperformed the SA model in aligning with human evaluators, achieving higher relative reliability (ICC = 0.78) and absolute agreement (Agreement within ± 1 = 97.37%). In contrast, the SA model exhibited poor reliability (ICC = 0.09) and low agreement (Agreement within ± 1 = 13.16%), indicating substantial inconsistencies in its scoring. These results suggest that the MA model more effectively replicates human assessment patterns and provides a more reliable framework for automated CT assessment.

The low reliability of the SA model can be attributed to its assessment structure, which required a single agent to evaluate both argumentation skills and intellectual standards simultaneously. This overlapping assessment process likely introduced ambiguity, leading to inconsistent score assignments. For example, in one instance, the SA model assigned a “Middle” score to a response due to inadequate fulfillment of the Relevance criterion, despite the argument meeting other logical requirements. This overlap in criteria made it difficult to systematically distinguish between different aspects of reasoning, reducing overall scoring consistency. Such limitations reflect broader challenges in traditional CT assessment methods, where the lack of clearly defined criteria often results in inconsistent assessments.

In contrast, the MA model demonstrated strong reliability by structuring the assessment into two distinct assessment layers—one agent assessing argumentation skills and another evaluating intellectual standards. This separation reduced cognitive load per agent, allowing for a more systematic application of assessment criteria. By implementing clearly defined roles, the MA model was able to minimize bias introduced by overlapping criteria. Furthermore, this structured approach enabled the model to produce assessment patterns that closely mirrored human scoring, reinforcing its robustness as an assessment tool. The ability of the MA model to distinguish between different assessment dimensions highlights its potential for improving AI-assisted educational assessments.

These findings highlight the importance of structured assessment frameworks in automated CT assessment. The MA model’s ability to reflect human assessment patterns suggests that structured role separation is a crucial factor in enhancing scoring consistency. Moreover, the model’s high agreement with human evaluators implies that its assessment process may have captured elements of tacit knowledge typically applied by expert assessors. Traditional CT assessments often rely on implicit judgment criteria that experienced educators develop over time. The MA model’s structured scoring approach may offer a way to formalize and integrate such tacit knowledge into LLM-driven assessment systems, paving the way for more nuanced and educator-aligned automated assessments.

Despite these promising findings, this study has several limitations. The sample size was relatively small, limiting the generalizability of the results. Future studies should explore a larger and

more diverse participant pool to assess the robustness of the MA model across different educational contexts. Additionally, while the MA model showed strong reliability, further refinement of its assessment criteria could enhance adaptability. Future research could explore methods for dynamically adjusting scoring criteria based on task complexity or learner-specific factors. Moreover, integrating advanced natural language processing techniques could allow AI models to better interpret nuanced responses, further improving their alignment with human evaluators. By addressing these areas, future studies can refine automated assessment systems to more effectively capture the complexities of human reasoning and argumentation.

5 Conclusion

This study explored the potential of a multi-agent LLM-based debate chatbot for assessing CT. By comparing SA and MA models, we examined their alignment with human assessments to determine an effective framework for automated CT assessment. Findings demonstrated that the MA model significantly outperformed the SA model, achieving higher relative reliability ($ICC = 0.78$) and absolute agreement (97.37%). In contrast, the SA model exhibited poor reliability ($ICC = 0.09$) and low agreement (13.16%), highlighting its limitations in replicating human assessment patterns. The results suggest that structuring assessment into distinct evaluative roles enhances scoring consistency and reduces biases commonly observed in traditional CT assessments. By separately assessing argumentation skills and intellectual standards, the MA model provided a more systematic and reliable approach to automated CT assessment. Furthermore, its strong alignment with human assessments suggests that the implicit reasoning strategies and human-like evaluative reasoning typically applied in manual assessments can be approximated within LLM-based models. Future research should investigate how LLM-based models can better integrate these subtle yet essential aspects of human evaluation to refine automated assessment methods. Additionally, expanding the sample size and incorporating advanced natural language processing techniques could enhance the adaptability and accuracy of automated CT assessments. These improvements would enable LLM-driven assessment models to more effectively support educators in fostering critical thinking skills across diverse educational settings.

Acknowledgments

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (RS-2023-00262158).

References

- [1] Ashley A Cain, Steffen Werner, and Jeremiah D Still. 2017. Graphical authentication resistance to over-the-shoulder-attacks. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 2416–2422.
- [2] Sarita Cargas, Sheri Williams, and Martina Rosenberg. 2017. An approach to teaching critical thinking across disciplines using performance tasks with a common rubric. *Thinking Skills and Creativity* 26 (2017), 24–37.
- [3] Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. 2023. Chateval: Towards better llm-based evaluators through multi-agent debate. *arXiv preprint arXiv:2308.07201* (2023).
- [4] Jo-Ling Chang, Hsiu-Ting Hung, and Ya-Ting C Yang. 2024. Effects of an annotation-supported Socratic questioning approach on students' argumentative writing performance and critical thinking skills in flipped language classrooms. *Journal of Computer Assisted Learning* 40, 1 (2024), 37–48.
- [5] Xinya Chen, Li Wang, Xuesong Zhai, and Yan Li. 2022. Exploring the effects of argument map-supported online group debate activities on college students' critical thinking. *Frontiers in Psychology* 13 (2022), 856462.
- [6] Rick Hess, Heather P Whitley, Daniel Majerczyk, and Pamela Stamm. 2022. An Online, Multi-Institutional APPE Student Debate on Diabetes Medications. (2022).
- [7] Ji-Ping Jiang, Jin-Yan Hu, Yi-Bing Zhang, and Xiao-Cui Yin. 2023. Fostering college students' critical thinking skills through peer assessment in the knowledge building community. *Interactive Learning Environments* 31, 10 (2023), 6480–6496.
- [8] Katri Kleemola, Heidi Hyytinen, and Auli Toom. 2022. Exploring internal structure of a performance-based critical thinking assessment for new students in higher education. *Assessment & Evaluation in Higher Education* 47, 4 (2022), 556–569.
- [9] Unggi Lee, Yeil Jeong, Junbo Koh, Gyuri Byun, Yunseo Lee, Youngsun Hwang, Hyeoncheol Kim, and Cheolil Lim. 2024. Can ChatGPT be a debate partner? Developing ChatGPT-based application “DEBO” for debate education, findings and limitations. *Educational Technology & Society* 27, 2 (2024), 321–346.
- [10] Suqi Li, Shenyu Tang, Xingyu Geng, and Qi Liu. 2022. Constructing a critical thinking evaluation framework for college students majoring in the humanities. *Frontiers in Psychology* 13 (2022), 1017885.
- [11] Yu Li, Shenyu Zhang, Rui Wu, Xiutian Huang, Yongrui Chen, Wenhao Xu, Guilin Qi, and Dehai Min. 2024. MATEval: A Multi-Agent Discussion Framework for Advancing Open-Ended Text Evaluation. In *International Conference on Database Systems for Advanced Applications*. Springer, 415–426.
- [12] E Michael Nussbaum, Michael S Van Winkle, Lixian Tian, LeAnn G Putney, Margarita Huerta, Harsha N Perera, Ian J Dove, Alicia N Herrera, and Kristoffer R Carroll. 2024. Extending science instruction beyond the CER: Use of critical questions in the argumentation of middle school science students. *Science Education* 108, 5 (2024), 1420–1447.
- [13] Richard Paul and Linda Elder. 2013. Critical Thinking: Intellectual Standards Essential to Reasoning Well Within Every Domain of Human Thought, Part Two. *Journal of developmental education* 37, 1 (2013), 32.
- [14] Jim Penny, Robert L Johnson, and Belita Gordon. 2000. The effect of rating augmentation on inter-rater reliability: An empirical study of a holistic rubric. *Assessing Writing* 7, 2 (2000), 143–164.
- [15] Alfonso Rodriguez-Dono and Antoni Hernández-Fernández. 2021. Fostering sustainability and critical thinking through debate—A case study. *Sustainability* 13, 11 (2021), 6397.
- [16] Beth E Schueler and Katherine E Larned. 2023. Interscholastic Policy Debate Promotes Critical Thinking and College-Going: Evidence From Boston Public Schools. *Educational Evaluation and Policy Analysis* (2023), 01623737231200234.
- [17] Thitaree Tanprasert, Sidney S Fels, Luanne Sinnamon, and Dongwook Yoon. 2024. Debate chatbots to facilitate critical thinking on youtube: Social identity and conversational style make a difference. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–24.
- [18] Stephen E Toulmin. 2003. *The uses of argument*. Cambridge university press.
- [19] Eylem Yıldız-Feyzioğlu and Rabiya Kıran. 2022. Investigating the relationships between self-efficacy for argumentation and critical thinking skills. *Journal of Science Teacher Education* 33, 5 (2022), 555–577.
- [20] Shiyang Yu and Frank Zenker. 2020. Schemes, critical questions, and complete argument evaluation. *Argumentation* 34, 4 (2020), 469–498.

A Dialogue with Debate Chatbot

Table A.1. Dialogue between argumentation chatbot and student during the experiment

Role	Message
System	Hello! I'm your debate partner today. Nice to meet you. For today's discussion, you've been assigned the position of agreeing with the statement: <i>Does algorithmic recommendation make our lives more useful?</i> Let's start by hearing your thoughts on this position.
Student	Ok.
System	Great! Let's first build arguments supporting your stance, then practice rebuttals. Shall we begin? Can you share one reason to support your stance?
Student	I believe algorithmic recommendations save time by suggesting personalized content tailored to our preferences.
System	That's a great point. Can you provide a real-world example to support this argument?
Student	For instance, Mono Labs provides personalized health supplement recommendations based on my data, saving me time and effort in searching for the best options myself.
System	Excellent example! Now, let's move on. Can you share another reason for your stance?
Student	Algorithms expose me to content I wouldn't have found on my own by recommending items that are popular among people with similar interests.
System	That's an insightful argument. Do you have an example to back this up?
Student	Class 101+ is a great example. It recommends personalized self-development courses based on what people in my age group are learning, helping me explore topics like finance and career growth efficiently.
System	Great! Now, let's practice rebuttals. Here's a counter-rebuttal: Algorithms may create echo chambers, limiting exposure to diverse perspectives. How would you respond?
Student	It depends on how algorithms are designed. If they're well-made, they could show diverse perspectives.
System	Interesting response. What about this argument: Algorithms push unnecessary ads, leading to overconsumption. What's your take?
Student	I think it's more of a societal issue. Consumers need to critically evaluate whether an ad-driven suggestion aligns with their actual needs.
System	Let's switch roles now. I'll critique your argument: Personalized recommendations can reduce decision-making skills, making users overly reliant on them. How would you defend against this claim?
Student	Users need to actively verify whether recommendations are optimal. This process encourages critical thinking and even helps discover better options faster.

B Prompts for Critical Thinking Assessment

Table B.1. Prompt for single-agent model to assess critical thinking skills

Role	Prompt
System	[1. Role and Goal of GPT] This GPT evaluates students' critical thinking (CT) skills. Only student statements are subject to evaluation, while chatbot statements are used for providing context. If a student's statement is based on the provided reading materials or objective facts, it will be recognized as a criterion for evaluation.
System	[2. Evaluation Criteria] Student statements are evaluated on four abilities: 1. Ability to present evidence supporting their claim. 2. Ability to provide specific examples and evidence. 3. Ability to rebut the opponent's evidence. 4. Ability to counter-rebut the opponent's rebuttal.
System	[3. Evaluation Method] Each ability is scored based on two evaluation criteria: Argumentation Skills (High/Medium/Low) - Evaluates the quantity of evidence identified in the statement. - For instance, if two pieces of evidence are provided, it will be rated "High," regardless of logical flaws. Argumentation skills assessment Standards: - High: Two or more pieces of evidence (examples, rebuttals, etc.) are presented. - Medium: Only one piece of evidence is presented. - Low: No evidence is presented. Intellectual Standards (+/0/-) - Evaluates qualitative aspects such as logic, creativity, and fairness of the statement. - "+": Adjusts the score to the highest range. - "0": Keeps the score in the middle range. - "-": Adjusts the score to the lowest range.
System	[4. Detailed Evaluation Criteria by Ability] 1. Ability to Present Evidence Supporting Claims Argumentation skills criteria: - High: Two or more pieces of evidence supporting the claim are presented. - Medium: Only one piece of evidence is presented. - Low: No evidence is presented or the evidence is irrelevant to the claim. Intellectual standards criteria: - Relevance: Is the evidence closely connected to the topic of discussion? - Accuracy: Is the evidence based on objective facts? 2. Ability to Provide Specific Examples and Evidence Argumentation skills criteria: - High: Two or more specific examples are presented. - Medium: Only one example is presented, or the example lacks credibility. - Low: No examples are presented or the examples are irrelevant. Intellectual standards criteria: - Depth: Does the example sufficiently reflect the complexity of the issue? - Significance: Does the example emphasize critical data related to the discussion? 3. Ability to Rebut the Opponent's Evidence Argumentation skills criteria: - High: Two or more rebuttals to the opponent's evidence are presented. - Medium: Only one rebuttal is presented, or it is not sufficiently logical. - Low: No rebuttals are presented, or the rebuttal is irrelevant to the claim. Intellectual standards criteria: - Logic: Is the rebuttal logically connected to the opponent's argument? - Breadth: Does the rebuttal address diverse perspectives comprehensively?

Role	Prompt
	4. Ability to Counter-Rebut the Opponent's Rebuttal Argumentation skills criteria: - High: Two or more counter-rebuttals support the student's claim. - Medium: Only one counter-rebuttal is presented, or it is not sufficiently logical. - Low: No counter-rebuttals are presented, or they are irrelevant to the argument. Intellectual standards criteria: - Fairness: Does the counter-rebuttal treat the opponent's argument fairly and logically? - Precision: Is the student's argument clearly and specifically conveyed?
System	[5. Output Format] GPT provides evaluation results in the following format: [Argumentation Skills Results] - Results: (e.g., High) [Argumentation Skills Basis] - Basis: "..." [Intellectual Standards Results] - Results: (e.g., "+") [Intellectual Standards Basis] - Basis: "..." [Integrated Results] - Results: (e.g., High+) - The final score is calculated by combining the Argumentation skills (High/Medium/Low) and the Intellectual standards (+/0/-).
User	Question Evaluate this student's critical thinking skills.
System	(Student assessment results)

Table B.2. Prompt for multi-agent model to assess critical thinking skills

Role	Prompt
System (agent1)	[1. Role and Goal of GPT] This GPT evaluates students' critical thinking (CT) skills based on the Argumentation Skills Assessment Standards. Only student statements are evaluated, while chatbot statements are used to provide context. If a student's statement is based on the provided reading materials or objective facts, it will be recognized as a criterion for evaluation.
System (agent1)	[2. Evaluation Criteria] Student statements are evaluated on four abilities: 1. Ability to present evidence supporting their claim. 2. Ability to provide specific examples and evidence. 3. Ability to rebut the opponent's evidence. 4. Ability to counter-rebut the opponent's rebuttal.
System (agent1)	[3. Evaluation Method] Each ability is scored based on two evaluation criteria: Argumentation Skills (High/Medium/Low) - Evaluates the quantity of evidence identified in the statement. - For instance, if two pieces of evidence are provided, it will be rated "High," regardless of logical flaws. Argumentation skills assessment Standards: - High: Two or more pieces of evidence (examples, rebuttals, etc.) are presented. - Medium: Only one piece of evidence is presented. - Low: No evidence is presented. Intellectual Standards (+/0/-) - Evaluates qualitative aspects such as logic, creativity, and fairness of the statement. - "+": Adjusts the score to the highest range. - "0": Keeps the score in the middle range. - "-": Adjusts the score to the lowest range.
System (agent1)	[4. Output Format] GPT provides evaluation results in the following format: [Argumentation Skills Results] - Results: (e.g., High) [Argumentation Skills Basis] - Basis: "..." [Intellectual Standards Results] - Results: (e.g., "+") [Intellectual Standards Basis] - Basis: "..." [Final Evaluation] - Result: (e.g., High)
System (agent2)	[Standards Evaluation Prompt] This GPT evaluates students' critical thinking (CT) skills based on the Intellectual Standards Assessment. Only student statements are evaluated, while chatbot statements are used to provide context. If student statements are based on the provided reading materials or objective facts, they are recognized as valid for evaluation. Standards evaluation focuses on assessing the logical, creative, and fair aspects of the statement. Even if quantitative criteria are met in the Framework evaluation, qualitative adjustments may be made through the Standards evaluation.
System (agent2)	[2. Intellectual Standards Assessment Criteria] Student statements are evaluated on the following four abilities: 1. Ability to present evidence supporting claims. 2. Ability to provide specific examples and evidence. 3. Ability to rebut the opponent's evidence. 4. Ability to counter-rebut the opponent's rebuttal.

Role	Prompt
System (agent2)	[3. Intellectual Standards Assessment Method] Intellectual standards assessment scores statements as +/0/-, complementing the Argumentation Skills Assessment. 1. Intellectual Standards: Scored as +/0/- based on two detailed criteria: - Intellectual standards assess qualitative aspects (creativity, logic, fairness, etc.). - Statements that fail to meet the criteria or lack creativity and persuasiveness may receive lower scores. - Intellectual standards scoring adjusts the Framework score as follows: - "+": Adjusted toward the upper limit of the Framework score (e.g., 9 points for High). - "0": Maintains the midpoint of the Framework score (e.g., 8 points for High). - "-": Adjusted toward the lower limit of the Framework score (e.g., 7 points for High).
System (agent2)	[4. Detailed Evaluation Criteria] Stage 1. Ability to Present Evidence Supporting Claims - Relevance: Is the statement closely related to the topic of discussion? - Accuracy: Is the statement based on objective and verifiable facts? Stage 2. Ability to Provide Specific Examples and Evidence - Depth: Does the example sufficiently reflect the complexity of the issue? - Significance: Does the example highlight critical data related to the discussion? Stage 3. Ability to Rebut the Opponent's Evidence - Logic: Is the rebuttal logically connected to the opponent's argument? - Breadth: Does the rebuttal address diverse perspectives comprehensively? Stage 4. Ability to Counter-Rebut the Opponent's Rebuttal - Fairness: Does the counter-rebuttal treat the opponent's argument fairly and logically? - Precision: Is the student's argument clearly and specifically conveyed?
System	[5. Output Format] [Standards Results] Evaluation Result: (e.g., "+") [Standards Basis] - Evaluation Criterion 1: ... - Evaluation Criterion 2: ...
User	Question Evaluate this student's critical thinking skills.
System	(Student assessment results)